

VENKATESHWARLU SARANGI, Ph.D

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PROFESSIONAL SUMMARY

Results-driven R&D Project Leader with a Ph.D. in Materials Science & Engineering and 4+ years of hands-on AI/ML project delivery experience. Proven track record leading end-to-end development of LLM fine-tuning, multimodal RAG systems, and production MLOps pipelines from PoC to scalable deployment. Skilled in Agentic AI frameworks, autonomous agents, and cross-functional team supervision while translating complex technical roadmaps for clients and stakeholders. Seeking to drive knowledge-based AI solutions and strategic project execution at the next level.

TECHNICAL SKILLS

Agentic AI, LLM orchestration, multi-agent systems, RAG pipelines, LoRA/QLoRA fine-tuning, MLOps, Docker, Kubernetes, CI/CD, AWS SageMaker, Python, project lifecycle management, stakeholder communication, PoC-to-production delivery, statistical modeling, model evaluation (ROUGE, AUROC) MLOps, LLMOps, Python, SQL, PyTorch, TensorFlow.

DOMAIN SPECIFIC SMALL LANGUAGE MODEL OPTIMIZATION & DEPLOYMENT

Capstone Project-1 (March, 2026)

- Production-Grade Knowledge Distillation (KD) Pipeline for Small Language Models (SLMs).
- Transferred knowledge from large teacher (Qwen2.5-7B-Instruct) to compact student (Qwen2.5-1.5B-Instruct) model, enhancing reasoning and planning via SFT, reward modeling, and RL methods like PPO and DPO
- Achieved comparable performance with 80% reduced inference compute (latency/memory), while synthesizing large-scale, high-quality data through rewriting, augmentation, and generation for pretraining, SFT, and RL stages.
- Production deployment: vLLM serving (OpenAI-compatible API), Docker containerization, Streamlit; implemented robust evaluation methodologies to assess model performance.
- Key metrics: Perplexity/BLEU convergence, TTFT (Time-To-First-Token), throughput vs. baseline teacher; integrated tool usage and agent capabilities for complex task solving.

Capstone Project-2 (April, 2026)

- Production-Grade Vision Language Model (VLM) Training & Serving Platform.
- Built multimodal SLMs for Computer Vision + NLP Applications, incorporating RL for foundation model enhancement and system 2 thinking with advanced decoding like MCTS and A.
- Delivered full-stack VLM platform for training/serving small-scale vision-language models, teaching models to use tools, interact with APIs, and build multi-agent systems for complex tasks.
- Production infrastructure: vLLM multi-modal serving (Qwen-VL/LLaVA), tensor parallelism, FP8 inference on H100; Docker + K8s orchestration.
- Key capabilities: Image-text alignment, zero-shot classification, cross-modal retrieval with CLIP-style embeddings; applied data synthesis and evaluation for pretraining, SFT, and RL.

WORK EXPERIENCE

DATA SCIENTIST & ML ENGINEER

Freelancer at XRadio Ltd, HK
(Aug, 2025 – March, 2026)

Deep Learning and Multimodal Systems for ECG Signal Intelligence

- Built an explainable deep learning model for 12-lead **ECG time-series data classification**, addressing challenges from non-stationary signals and inter-patient variability.
- Evaluated multiple deep learning(CV) and vision architectures including ResNet34, ResNet50, DenseNet, VGG16, Vision Optimized Transformers (ViT) and Vision-Language Models (VLMs), multimodal AI systems for healthcare prediction tasks.
- Achieved strong model performance with ResNet34, reaching **AUROC 0.98 and F1-score 0.826** across nine arrhythmia categories.

DATA SCIENTIST & RESEARCH ENGINEER

Gense Technologies Ltd, HK
(Oct, 2022 – Aug, 2025)

Projects & Technical Experience

- Designed and deployed end-to-end supervised learning pipelines with ML models for clinical prediction tasks, achieving 87% sensitivity and >99% specificity on imbalanced biomedical datasets.
- Engineered feature extraction and signal processing workflows using Python (NumPy, Pandas, Scikit-learn) to improve SNR and diagnostic quality in time-series bio-conductivity data.
- Achieved a 100% improvement in skin electrode contact impedance without conductive gels, improving signal sensitivity.

- Developed and validated signal enhancement and impedance analysis models for ECG and EIT datasets, improving signal-to-noise ratio and diagnostic reliability.
- Built automated MLOps workflows with Docker, CI/CD, and MLflow for experiment tracking, model versioning, and reproducible deployments across cloud environments. Applied statistical validation frameworks (k-fold cross-validation, ROC-AUC optimization, hypothesis testing) to ensure diagnostic-grade model reliability and regulatory compliance.
- Developed synthetic data augmentation strategies (SMOTE, bootstrapping) to address limited clinical datasets, improving model generalization by 20%.
- Proficient in Python, SQL, PyTorch, Scikit-learn, and cloud ML platforms (GCP, Firebase, AWS SageMaker) for scalable production systems.

DATA SCIENTIST AND COMPUTATIONAL ANALYST

Anvipro IT Solutions, India

(December, 2021 – June, 2022)

Materials Science & Computational Modeling

- Applied supervised learning and statistical modeling techniques to analyze material properties, identify trends, and support predictive insights.
- Conducted physics-based simulation and data analysis using COMSOL Multiphysics to study material flexibility (>90%), stress distribution, hardness, and structural behavior.
- Performed materials data preprocessing, cleaning, and refinement using MATLAB and Python basic libraries to improve analysis quality and model readiness.
- Improved simulation and analytical workflows through iterative model refinement, results evaluation, and reliability analysis for engineering and materials-focused projects.
- **Technologies:** Python, Data processing, Feature Engineering, EDA, NumPy, Pandas, Scikit-learn, OriginLab, SQL, Power BI, & Statistical Modeling.

PH.D RESEARCH SCHOLAR

CityU HK

(August, 2015 – October 2021)

Materials Engineering / Computational Modeling / Data Analysis

- Architected piezoelectric electrodes and integrated AI-driven computational modeling for material structure analysis using X-ray and neutron diffraction datasets, aligning with predictive AI solutions for operational efficiency.
- Delivered Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning to extract structural insights from complex datasets, incorporating Machine Learning frameworks like Scikit-learn for enhanced prediction tasks.
- Published research in Nature Communications, Communications Physics, Physical Review B, Scientific Reports, and other high-impact journals, focusing on innovative AI applications in materials science.
- Architected piezoelectric electrodes **Large electrostrains of 0.2%** and integrated AI-driven computational modeling for material structure analysis using X-ray and neutron diffraction datasets, aligning with predictive AI solutions for operational efficiency.
- Delivered Rietveld refinement and data processing, feature engineering, and regression-based parameter optimization, fine-tuning methods, hyperparameter tuning **Goodness of Fit Chi-squared < 1%** alongside the **Weighted Profile R-factor (R_{wp}) < 3%**, to extract structural insights from complex datasets, incorporating Machine Learning frameworks like Scikit-learn for enhanced prediction tasks.

JOURNAL PUBLICATIONS

- S. Venkateshwarlu et al., **Nature Communications Physics**, (2020).
- S. Venkateshwarlu et al., **Journal of Materials Research**, Cambridge University Press JMR-0830 (2019).
- S. Nayak, S. Venkateshwarlu, et al., **Journal of the American Ceramic Society**, (2019).
- Frederick, Sarangi Venkateshwarlu, et al., **Chemistry of Materials** (2021).
- A. Pramanick, S. Venkateshwarlu, et al., **Physical Review B**, (2021).
- Liang, Zhuoxin Liu, S. Venkateshwarlu, et al., **Nature, Light: Science & Applications**, (2021).
- G. Srinivas, S. Venkateshwarlu, et al., **Applied Physics Letters**, (2015).
- S. Venkateshwarlu, et al., **Journal of Modern Materials**, (2016).
- Pramanick, Venkateshwarlu, **Journal of the European Ceramic Society**, (2023).
- Nirmal and Sarangi Venkateshwarlu et al., **arXiv:2508.10940**, (2025).
- Kwok, W.C., Sarangi Venkateshwarlu et al., **Nature Scientific Reports** (2025).

EDUCATION

IIT Bombay India and CityU HK,

- **Ph.D.** Materials Science and Eng (Physics), City University of Hong Kong, (Oct, 2021)

- **M.Tech.** Materials Science and Eng (Physics), IIT Bombay, India, (June, 2013)

TRAINING | CERTIFICATIONS

IIT Bombay India and CityU HK,

- **Generative AI with Large Language Models (LLMs)**
- DeepLearning.AI & Amazon Web Services | Coursera | February 2026
- Skills: Transformers Architecture (FLAN-T5, BERT, GPT, DeepSeek, Qwen, Phi-2, Gemma) · RL models · PyTorch · PEFT · Fine-tuning · LoRA · QLoRA · Knowledge Distillation(KD) · RAG · LangChain · LangGraph · Deployment · vLLM · OpenWebUI · FastAPI · Streamlit · RunPOD · LightningAI · HuggingFace, · Optuna Optimization
- **Machine Learning Specialization** | August 2025
- **Deep Learning and NLP Specialization** | January 2026
- Stanford University & DeepLearning.AI | Coursera
- **End-to-End MLOPS Bootcamp** | Udemy | April 2026-(On-going)
- **AWS Certified Cloud Practitioner** | Udemy | April 2026-(On-going)